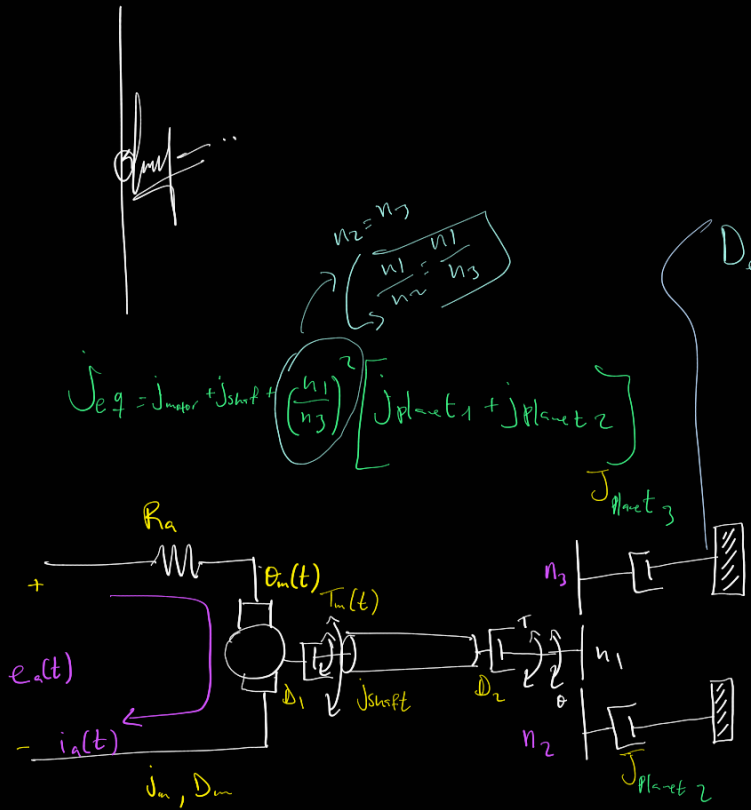
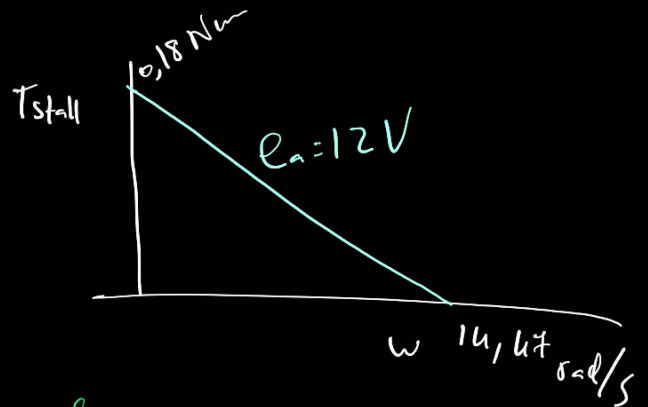


Mechanical Modeling



$$D_{eq} = D_m + D_{shaft} + \left(\frac{n_1}{n_2}\right)^2 [D_{planet2} + D_{planet3}]$$

$$J_{eq} = J_{motor} + J_{shaft} + \left(\frac{n_1}{n_3}\right)^2 [J_{planet1} + J_{planet2}]$$



for motor

$$12V$$

$$12mm \text{ (redüktör kutusu genişliği)}$$

$$100 \text{ rpm}$$

J_{shaft} = inertia of the long green shaft
+ sun gear (n_1)

$$D_{shaft} = D_1 + D_2$$

$$J_{load} = J_{planet3} + J_{planet2}$$

$$n = 100 \text{ dev/dk}$$

$$\omega = n \cdot \frac{2\pi}{60} = 100 \cdot \frac{2\pi}{60} = 10,47 \text{ rad/s}$$

$$T_{nominal} = 0,5 \text{ kg.cm} \sim 0,05 \text{ Nm}$$

$$T_{stall} = 1,8 \text{ kg.cm} \sim 0,18 \text{ Nm}$$

$$T_m(t) = J_{eq} \cdot \frac{d^2 \theta_m(t)}{dt^2} + D_{eq} \frac{d\theta_m(t)}{dt}$$

Electrical Modeling
(this models the N20 motor itself)

$V(t)$ - input voltage from L298N driver.

$i(t)$ - armature current

R - armature Resistance

L - armature inductance (negligible for N20 like small motors, but we include for accuracy).

k_b - Back EMF Constant

$k_t \rightarrow$ torque constant

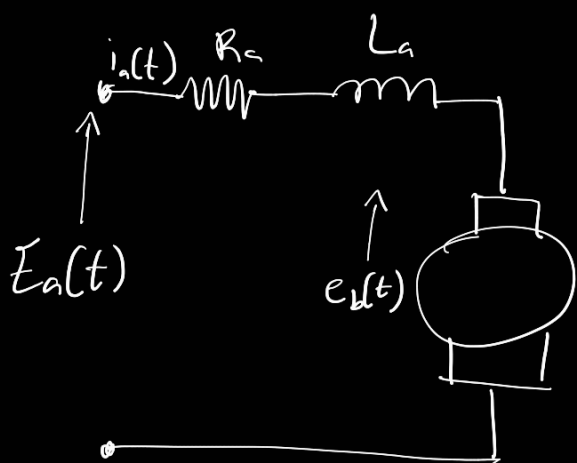
$$V(t) = R \cdot i(t) + L \frac{di(t)}{dt} + e_b(t)$$

$$e_b(t) = k_b \frac{d\theta_m(t)}{dt} \text{ or } k_b \cdot \omega$$

$\hookrightarrow$ inductor's voltage

$$T_m(t) = k_t \cdot i(t)$$

$\hookrightarrow$ electro-mechanical coupling



NICE
★

Block Diagram

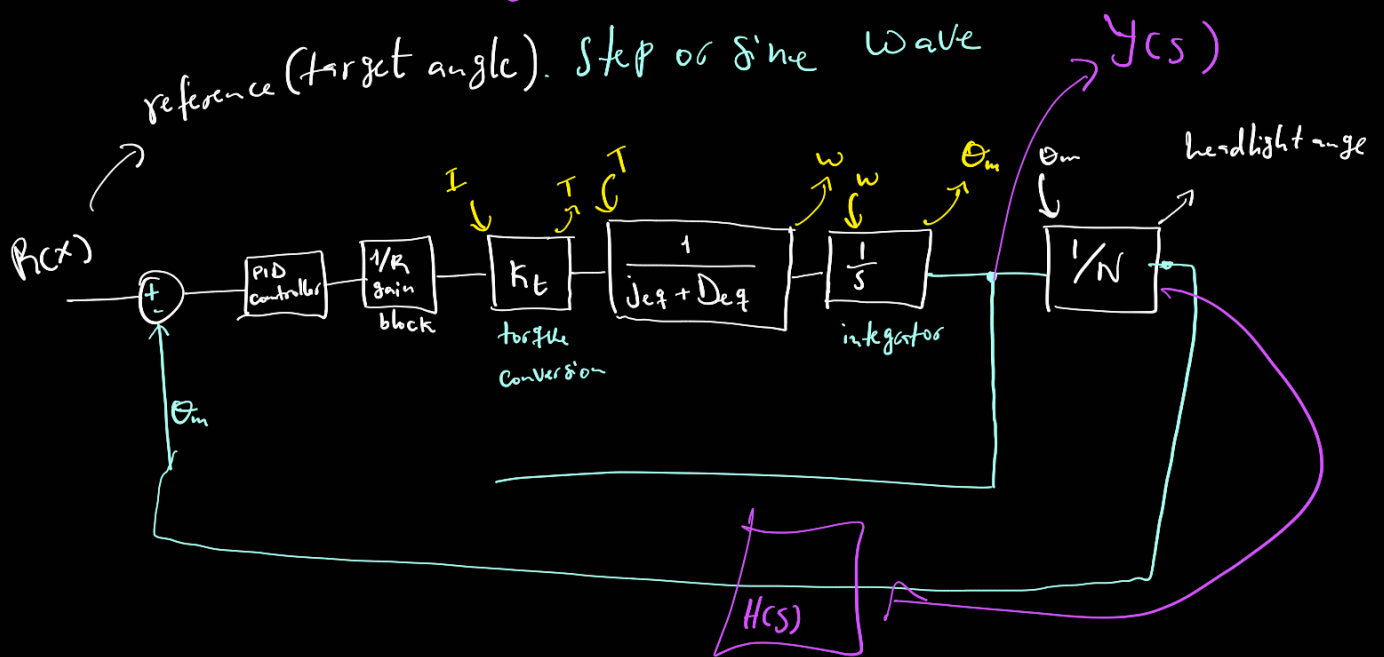
Transfer function of the DC motor

input $\rightarrow V_o / k_g$
 output $\rightarrow \text{Angle}$ } assuming $L = 0$ (common for NZO motors to simplify)

$$G(s) = \frac{\Theta(s)}{V(s)}$$

$$\frac{\Theta_m(s)}{V(s)} = \frac{k_t}{s \cdot [(J_s + D) + k_t k_b]}$$

block diagram Structure



Error = trim pot value - actual motor position.

$H(s) = 1$ (if pulses are converted into angle in code)

$H(s) = k_{enc}$ (if system works with the pulses)

$$k_{enc} = \frac{600 \text{ darbe}}{2\pi \text{ rad}} = 95,49 \text{ darbe/rad}$$